

Assignment: Nested Loops and Pattern Programs in C

1. Print a 5x5 Instagram Post Feed Using 📷 Symbols

```
#include <stdio.h>
```

```
int main()
{
    int i, j;

    for(i = 1; i <= 5; i++)
    {
        for(j = 1; j <= 5; j++)
        {
            printf("📷 ");
        }
        printf("\n");
    }

    return 0;
}
```

Sample Output:

```
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
```

2. Right-Angled Triangle Pattern Using Numbers

```
#include <stdio.h>

int main()
{
    int i, j;

    for(i = 1; i <= 5; i++)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }

    return 0;
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

3. Centered Pyramid Pattern with 6 Rows

```
#include <stdio.h>
```

```
int main()
{
    int i, j;

    for(i = 1; i <= 6; i++)
    {
        for(j = 1; j <= 6 - i; j++)
        {
            printf(" ");
        }

        for(j = 1; j <= (2 * i - 1); j++)
        {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

Output:

```
 *
***
*****
```

4. Alternating 0s and 1s Pattern (4x4 Grid)

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```
    for(i = 0; i < 4; i++)
```

```
    {
```

```
        for(j = 0; j < 4; j++)
```

```
        {
```

```
            printf("%d ", (i + j) % 2);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

0 1 0 1

1 0 1 0

0 1 0 1

1 0 1 0

5. Pyramid Pattern with User Input for Number of Rows

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int rows, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &rows);
```

```
    for(i = 1; i <= rows; i++)
```

```
    {
```

```
        for(j = 1; j <= rows - i; j++)
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
        for(j = 1; j <= (2 * i - 1); j++)
```

```
        {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
}
```

```
    return 0;  
}
```

Sample Input:

5

Sample Output:

```
  *  
  
 ***  
  
*****  
  
*****  
  
*****
```

Conclusion

This assignment demonstrated the use of nested loops to create different patterns such as grids, number triangles, centered pyramids, alternating binary patterns, and user-defined pyramid structures in C programming.